MY VISION

AI that keeps learning after training, on your own device.

Today's models are trained once, then frozen.
What they learn in a conversation is gone when it ends.

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THE IDEA

Learn like a brain

DAY

Fast memory

writes new experience in seconds

→

NIGHT

Sleep

replays its own dreams to keep what it knew

→

FOR LIFE

Long-term weights

hold old and new skills together

No old data is stored. It rehearses by dreaming.

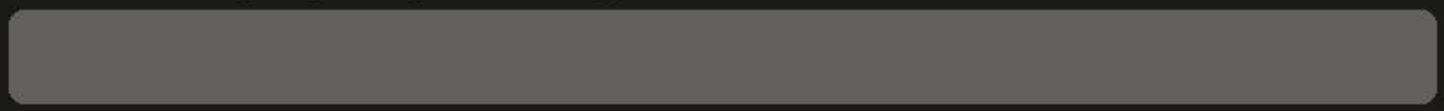
Inspired by complementary learning systems theory (hippocampus + cortex)

MOST PROMISING FINDING

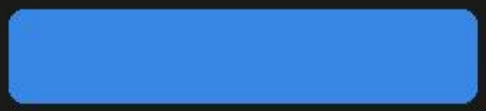
Up to 7× less forgetting

A small GPT learned 6 kinds of English, one per day. 3 seeds each.

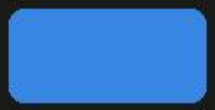
Normal day-by-day training



+ sleep on its own dreams



+ stronger sleep (more dreams, keep harder)



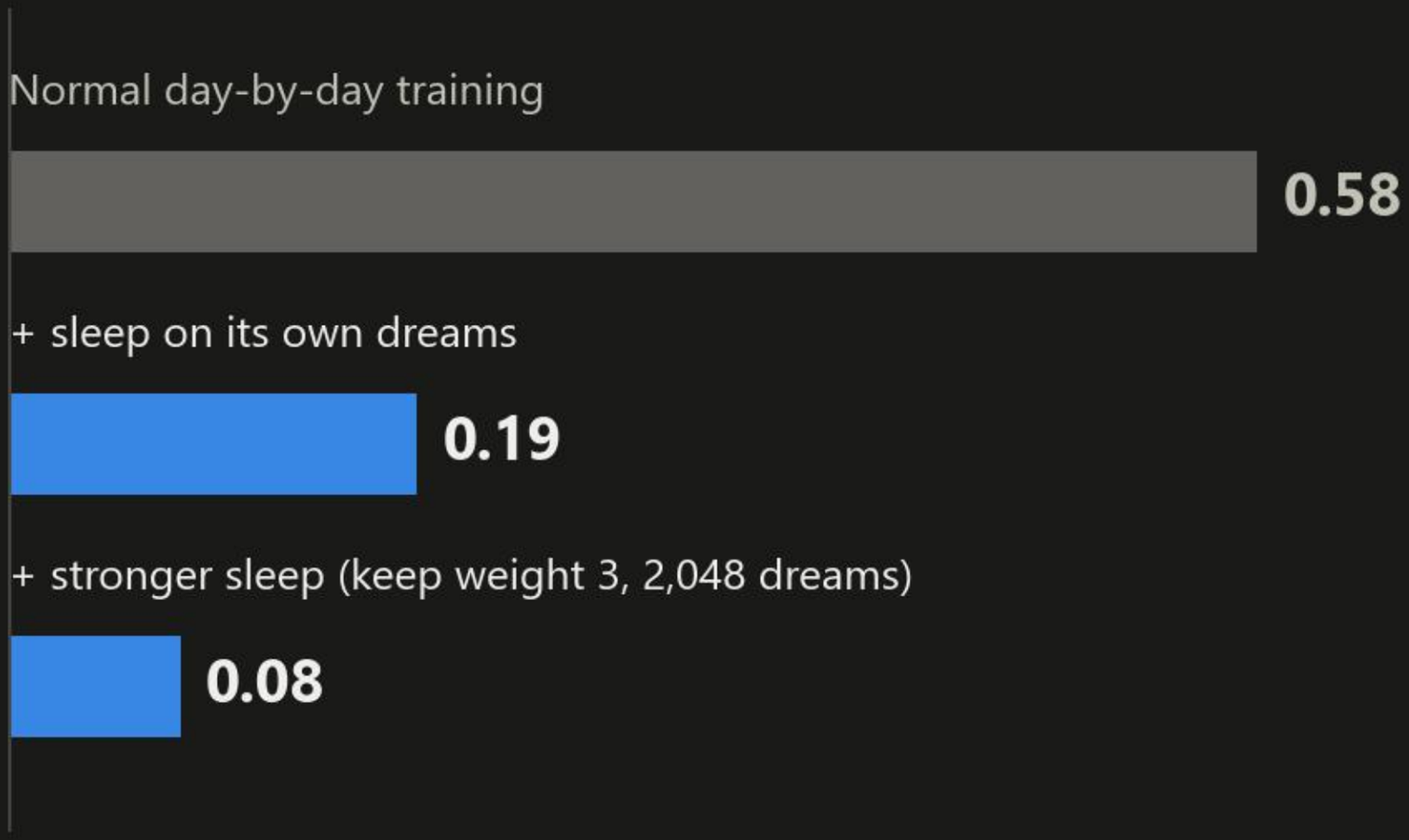
Forgetting in bits per byte, lower is better.

No old text is stored. It rehearses by dreaming.

Byte-level nanoGPT · BabyLM 10M, 6 domains · single RTX 4080 SUPER

Sleep cut forgetting 3–7×

A small GPT learns day by day. Between days it 'sleeps': keeps its old predictions on its own dreams.

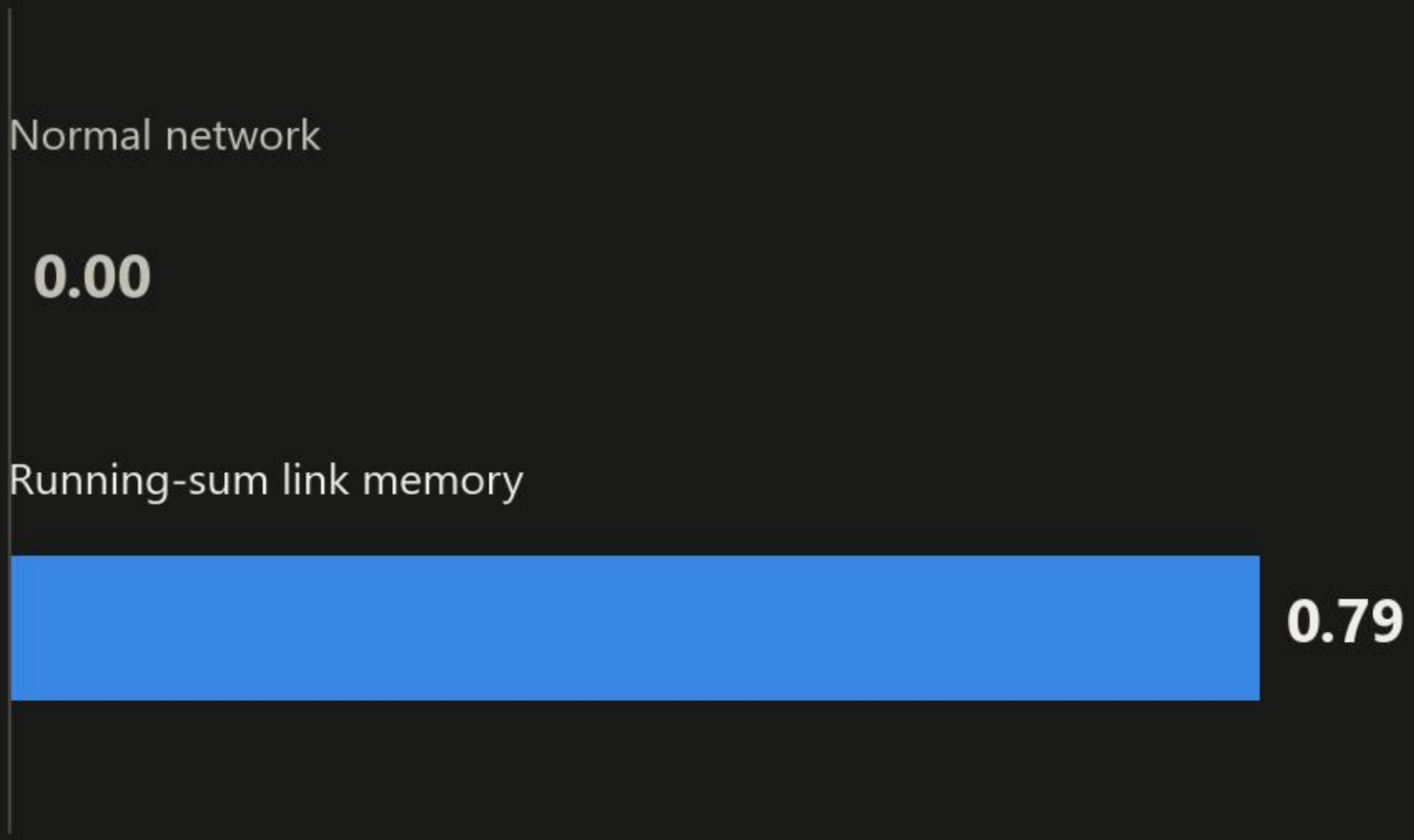


Forgetting in bits per byte, lower is better; 3 seeds per arm. The stronger setting was the best of 4 pre-defined arms, picked after the run. Much less forgetting, but not zero.

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Remembering day 1 after 20 days

CIFAR-100 learned as a 20-day stream, frozen image encoder. Accuracy on day-1 classes at the end.

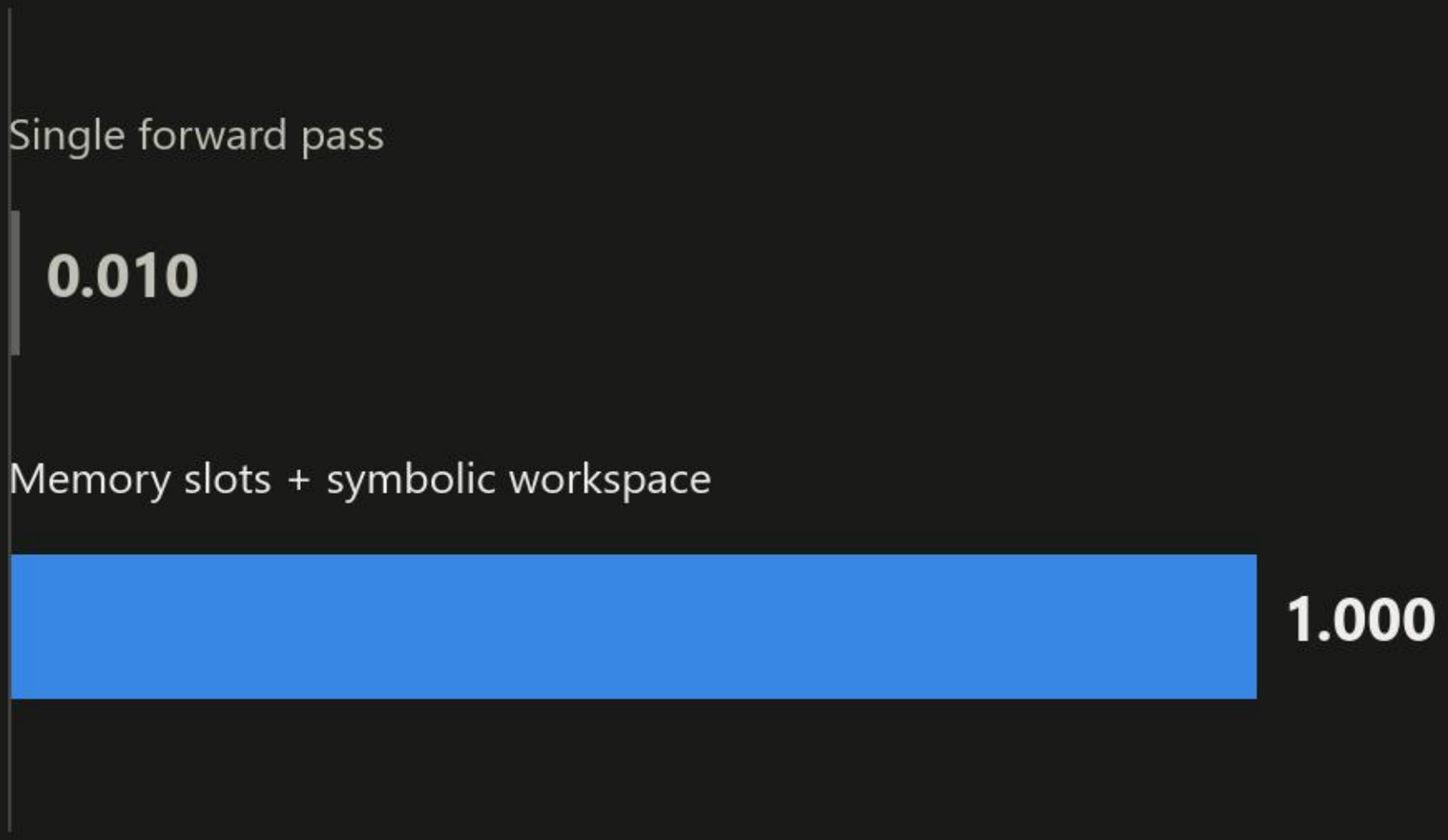


The link memory also matched learning everything at once on all classes (0.60 vs 0.60).

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Memory slots on a frozen transformer

Write a new fact in ~25 steps (~1 s) without touching the backbone, then answer 2-hop questions.

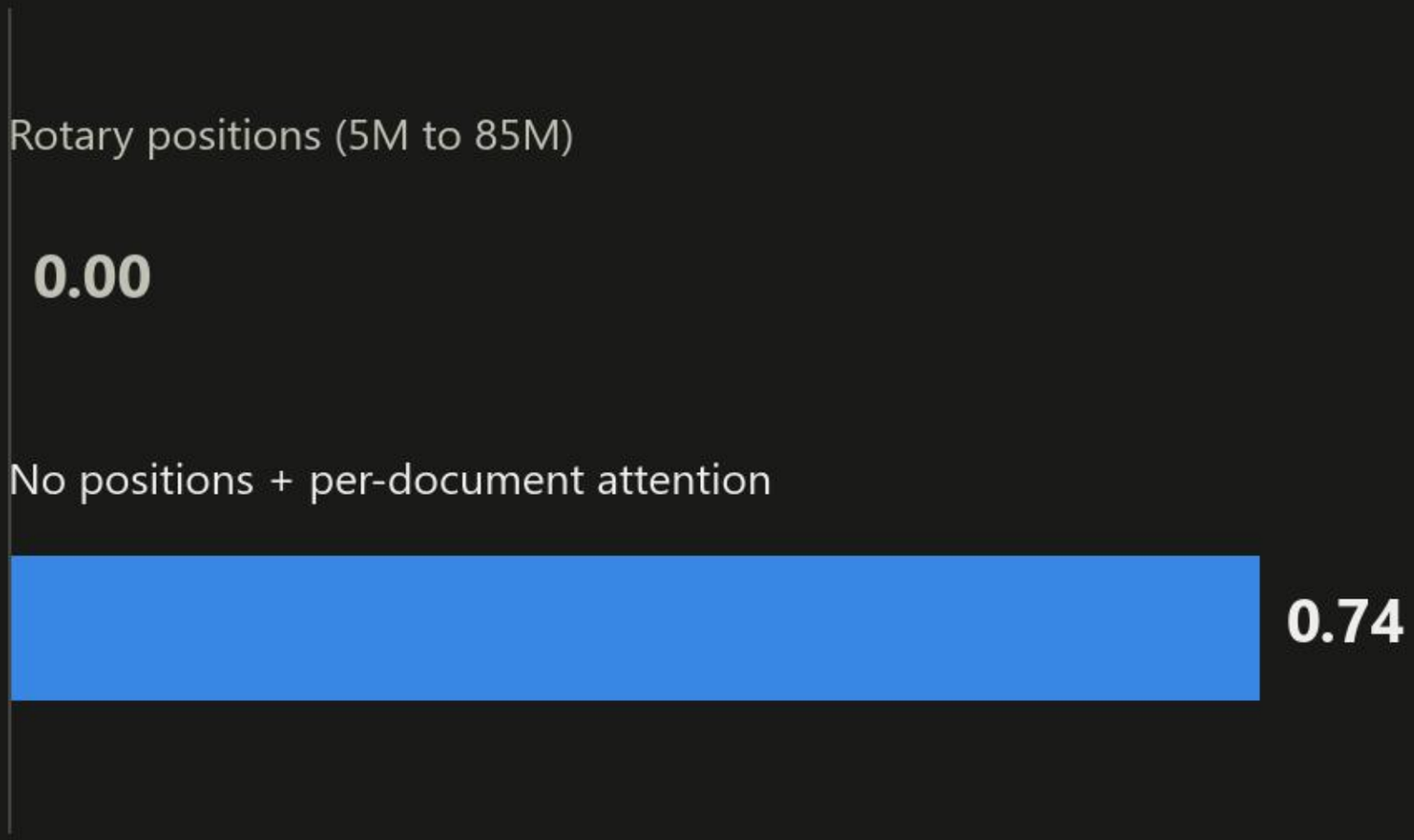


2-hop accuracy. 2,000 earlier facts stayed at 1.000, zero drift over 100 writes. Synthetic, closed vocabulary: unseen words still fail.

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Counting past what it was trained on

Small GPTs trained from scratch in a mixed prose + reasoning stream.
Test: 6-digit addition, longer than training.

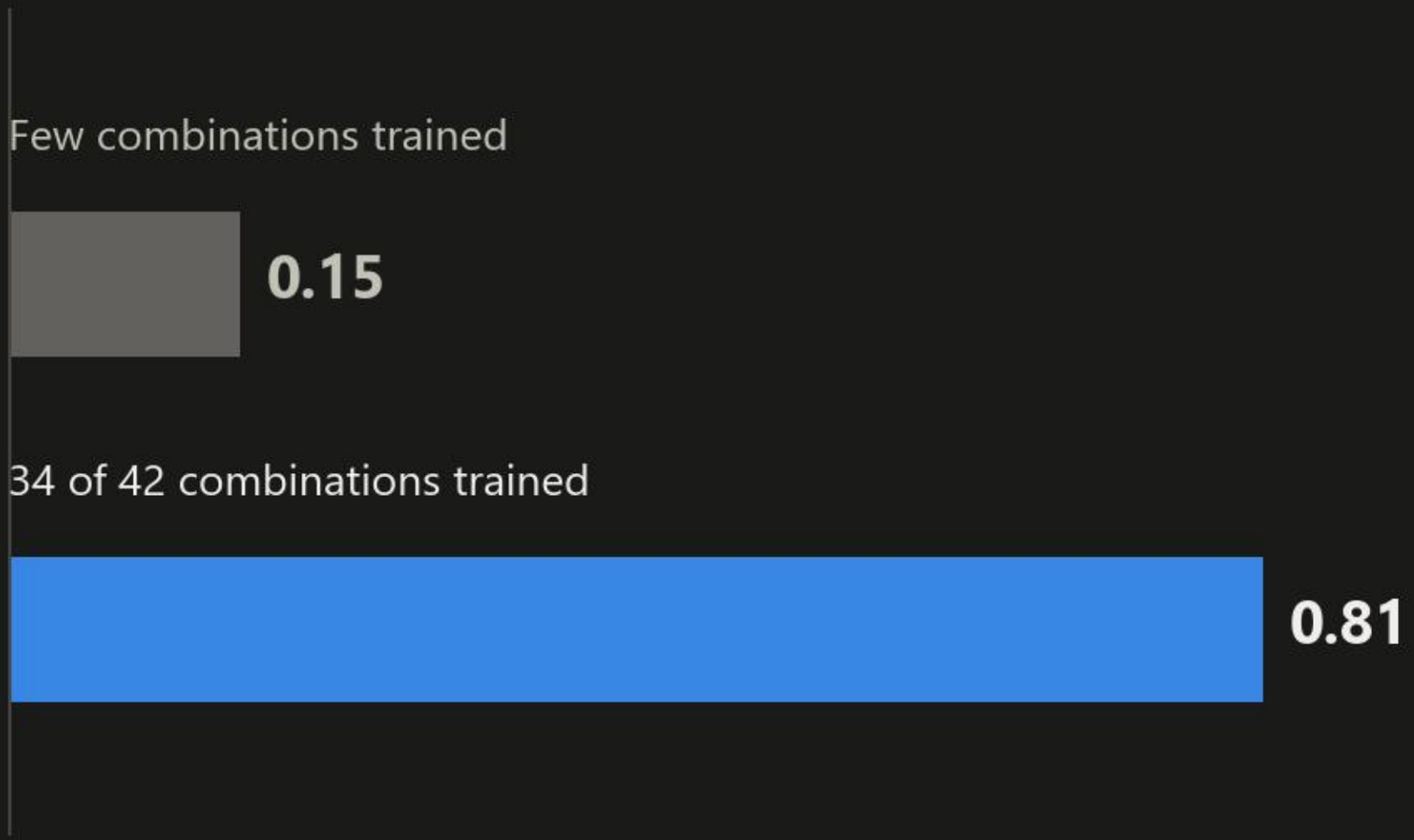


Scaling 5M → 85M left rotary at exactly 0.00. Removing positions and masking attention per document fixed it.

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Composition is a data problem

Can a 5M model chain two skills it never saw chained, when asked in English?

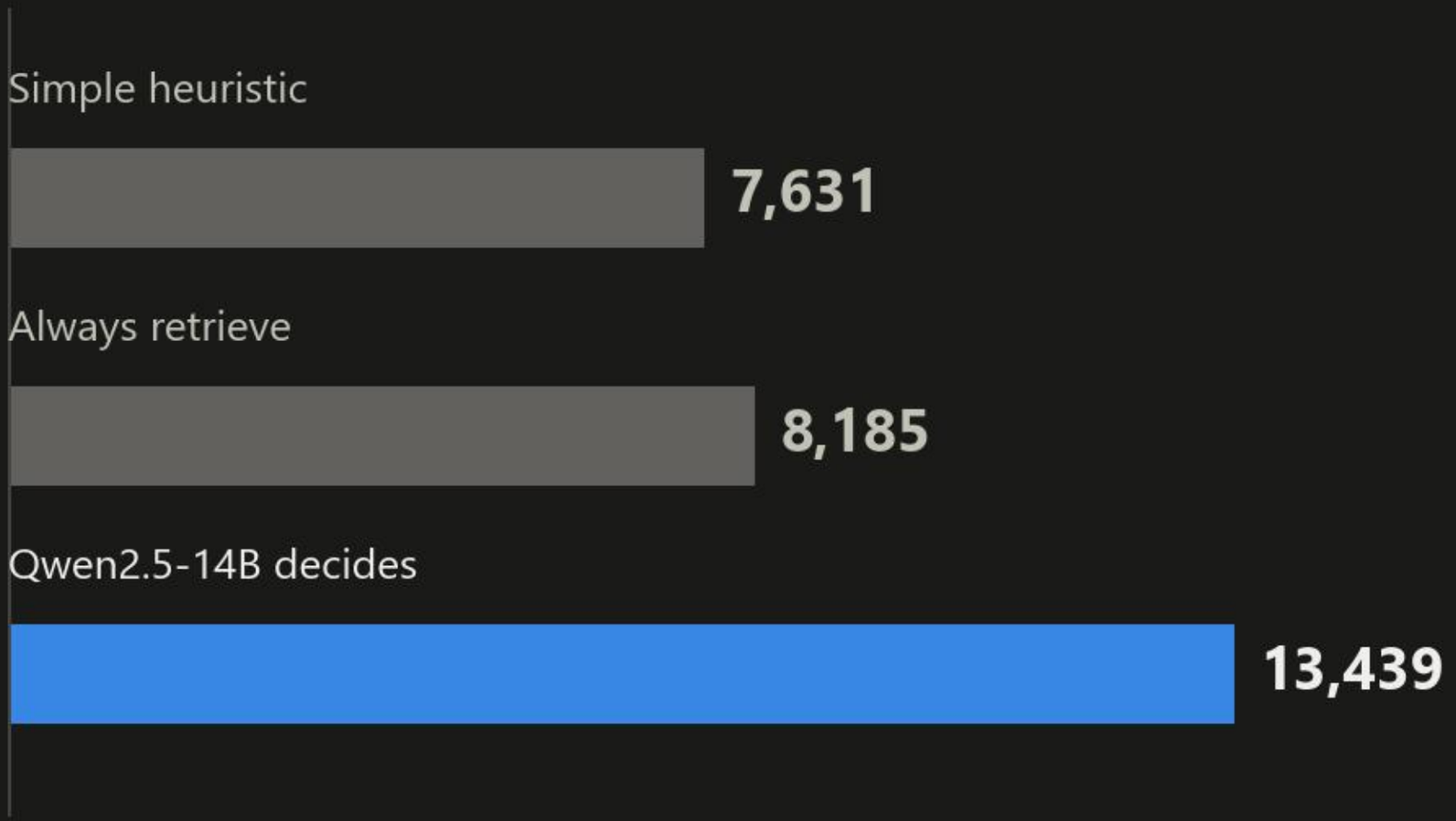


Accuracy on held-out pairs (low-diversity run: 0–15 of 100). More model size didn't help. More variety did.

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A 14B model as memory gate

Qwen2.5-14B (local) as a retrieve-or-act gate vs simple policies, same tasks.

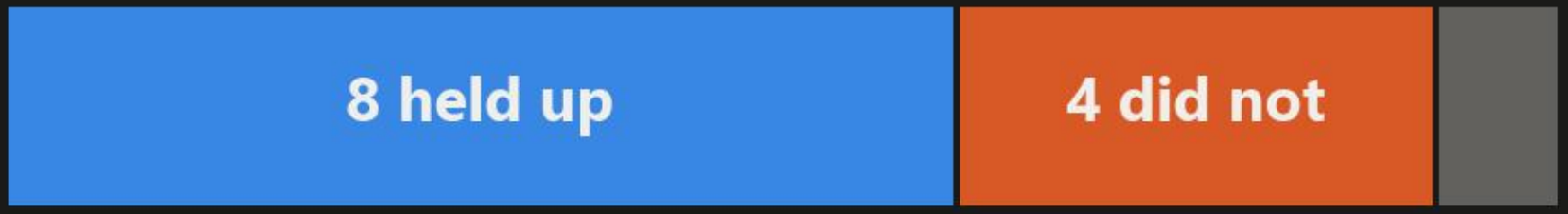


Total tokens, lower is better. The model gate also skipped two retrievals it needed. A negative result, logged as one.

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Re-testing a Nature paper myself

13 claims from Dohare et al., "Loss of plasticity in deep continual learning" (Nature 2024) and a Rich Sutton interview. Permuted MNIST, 3 seeds, falsifier written before each run.



1 undecided

Held up, e.g.

- Plain backprop loses plasticity
- Adam is worse
- L2 helps but doesn't eliminate it (−2.4 to −3.0 pts)
- Replay reduces forgetting

Did not hold in my setup, e.g.

- Shrink-and-perturb nearly eliminates the loss
- "Forgetting is totally curable" with continual backprop
- The three correlates all move as described

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NEXT

Make it last.

Over long streams, shared weights still forget. I'm building architectures where new learning doesn't overwrite old knowledge, small enough for a local device.

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github.com/LamShiuChing/ai-research-log-and-idea

20+ experiments · code · logs · papers